

Second-order Trotter error for a rigid rotor under white-noise spherical diffusion with dipolar, J and CSA couplings: a cluster-aware splitting

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Abstract

We estimate the discretization error of a second-order (Strang) circuit for the orientation-averaged spin dynamics of a rigid molecule undergoing isotropic rotational diffusion. The nuclear spins couple through dipolar interactions, isotropic and anisotropic J couplings, and isotropic and anisotropic chemical shifts. We work in the white-noise (motional-narrowing) regime, where the integrated rank-2 fields become independent Gaussian kicks, one per noise channel. The splitting is organised by geometric clusters of the molecule.

The averaged circuit is weak order 2 in every splitting except one: splitting a single-channel kick across clusters with a common Gaussian variable is only weak order 1. Its error generator is $\delta t \mathcal{G}$ with $\mathcal{G} = \frac{3}{2} \sum_{\mu} \gamma_{\mu}^2 \{\text{ad}_{W_{\mu}}, \text{ad}_{E_{\mu}}\}$, which is suppressed by the ratio ε_{geo} of inter- to intra-cluster couplings. A three-stage triple jump in the noise variable restores weak order 2; we show that no cheaper modification exists. All claims are checked numerically on a two-cluster $^{13}\text{C}_2$ fragment.

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1 Setup

The target is the orientation-averaged spin dynamics of a rigid molecule with N spin- $\frac{1}{2}$ nuclei. Its orientation $R(t) \in SO(3)$ performs isotropic Brownian motion, with white-noise angular velocity

$$\langle \omega_{\alpha}(t) \omega_{\beta}(t') \rangle = 2D_R \delta_{\alpha\beta} \delta(t - t'). \quad (1)$$

Rank- l Wigner functions then decay as $e^{-l(l+1)D_R t}$. The rank-2 correlation time is $\tau_c = 1/(6D_R)$.

Hamiltonian. The lab-frame Hamiltonian splits into a rotation-invariant part and a rank-2 part driven by R :

$$H(R) = \underbrace{\sum_i \omega_i I_{iz} + 2\pi \sum_{i < j} J_{ij} \mathbf{I}_i \cdot \mathbf{I}_j}_{H_0 \text{ (isotropic CS, isotropic } J)} + \underbrace{\sum_{a,b=1}^5 D_{ab}(R) W_{ab}}_{H_1(R) \text{ (DD, CSA, anisotropic } J)}, \quad (2)$$

with $\omega_i = -\gamma_i B_0(1 - \sigma_i^{\text{iso}})$. The rank-2 part uses

$$D_{ab}(R) = \text{tr}(E_a R E_b R^T), \quad W_{ab} = \sum_{\lambda} t_{\lambda b} Q_a^{\lambda}, \quad t_{\lambda b} = \text{tr}(E_b A_{\lambda}), \quad (3)$$

$$Q_a^{(ij)} = \sum_{\alpha\beta} (E_a)_{\alpha\beta} I_{i\alpha} I_{j\beta}, \quad Q_a^{(i,\text{CSA})} = \omega_i^0 \sum_{\alpha} (E_a)_{\alpha z} I_{i\alpha}. \quad (4)$$

- E_a ($a = 1, \dots, 5$) is a real orthonormal basis of traceless symmetric 3×3 matrices, with $\text{tr}(E_a E_b) = \delta_{ab}$.
- λ runs over interactions: dipolar pairs (ij) and the CSA of each spin i . A_{λ} is the molecular-frame spatial tensor.
- Dipolar: $A_{ij} = -b_{ij} (3\hat{n}_{ij}\hat{n}_{ij}^T - \mathbb{I})$, with $b_{ij} = \mu_0 \gamma_i \gamma_j \hbar / (4\pi r_{ij}^3)$.
- The anisotropic J coupling has the same spin tensor as the dipolar term of the same pair. It simply adds to A_{ij} (a pseudo-dipolar shift) and needs no separate treatment.
- CSA: A_i is the traceless symmetric part of σ_i . The antisymmetric (rank-1) part rotates with $D^{(1)}$, with correlation time $3\tau_c$. It adds three channels of the same form; we drop it, but the analysis applies unchanged.

Correlations. Isotropy gives

$$\mathbb{E}[D_{ab}(R(0)) D_{cd}(R(t))] = \frac{\delta_{ac}\delta_{bd}}{5} e^{-|t|/\tau_c}. \quad (5)$$

One rotation drives every interaction, so every pair (λ, λ') is cross-correlated with weight $t_{\lambda} \cdot t_{\lambda'}$. For axial tensors this is $\propto P_2(\cos \theta_{\lambda\lambda'})$, where $\theta_{\lambda\lambda'}$ is the angle between principal axes. This includes interactions in distant parts of the molecule, which is why the cluster splitting below must use one common noise per channel.

Scales. Table 1 lists typical sizes at 14.1 T and $\tau_c = 100$ ps. With $h \approx 2\pi \times 25$ kHz, the motional-narrowing parameter is $\kappa = h\tau_c \approx 1.6 \times 10^{-5}$, and relaxation rates are $\Gamma \sim h^2 \tau_c \sim 1 \text{ s}^{-1}$.

Interaction	Size/ 2π
$^{13}\text{C}-^1\text{H}$ dipolar, 1.09 Å	23 kHz
$^1\text{H}-^1\text{H}$ geminal dipolar, 1.78 Å	21 kHz
$^1\text{H}-^1\text{H}$ between groups, 2.5–3 Å	4–8 kHz
^{13}C CSA, $\Delta\sigma \approx 150$ ppm	23 kHz
^1H CSA, $\Delta\sigma \approx 5$ ppm	3 kHz
$^1J_{\text{CH}} / ^3J_{\text{HH}}$	140 Hz / 2–10 Hz

Table 1: Typical interaction strengths.

2 Tensor structure, frequency channels and the white-noise limit

Only H_1 carries the rotation; H_0 is invariant and deterministic. In the rotating frame of the Zeeman terms, each Q_a^λ splits into eigenoperators of ad_{H_Z} with Bohr frequency ν ($0, \pm\omega_I, \pm(\omega_I \pm \omega_S), \pm 2\omega_I, \dots$). The noise channels are therefore labelled $\mu = (a, b, \nu)$:

$$W_\mu = \sum_\lambda t_{\lambda b} Q_{a,\nu}^\lambda, \quad [H_Z, W_\mu] = \nu W_\mu. \quad (6)$$

There are two regimes for the step size δt .

- **Resolved** ($\delta t \ll \tau_c, 1/\omega_0$). Sample $R(t_k)$ on the grid with exact or curvature-corrected $SO(3)$ increments, and evaluate $H(R_k)$ at step midpoints. This needs about 10^{11} steps per second of evolution, so it serves only for validation.
- **White-noise** ($\tau_c \ll \delta t, \Gamma \delta t \ll 1$). Over one step the orientation decorrelates many times, since $\kappa \sim 10^{-5}$. The integrated fields become Gaussian kicks, one per channel, with diagonal covariance:

$$\Xi_\mu = \int_{t_k}^{t_k + \delta t} e^{i\nu t} D_{ab}(R(t)) dt, \quad \mathbb{E}[\Xi_\mu \Xi_{\mu'}^*] = \delta_{\mu\mu'} \gamma_\mu \delta t, \quad \gamma_\mu = \frac{\tilde{J}(\nu)}{5} = \frac{2\tau_c}{5(1 + \nu^2 \tau_c^2)}. \quad (7)$$

Each channel's kick averages to its own dissipator. For a real ($\nu = 0$) channel, with $X \sim \mathcal{N}(0, \gamma_\mu \delta t)$,

$$\mathbb{E}_X[e^{-iX \text{ad}_{W_\mu}}] = e^{\delta t \mathcal{D}_\mu}, \quad \mathcal{D}_\mu = -\frac{\gamma_\mu}{2} \text{ad}_{W_\mu}^2. \quad (8)$$

A circular complex channel ($\pm\nu$) is split into two independent Hermitian kicks. The target generator is secular Redfield plus the dynamic frequency shift (DFS):

$$\mathcal{L} = -i \text{ad}_{H_0 + H_{\text{DFS}}} + \sum_\mu \mathcal{D}_\mu. \quad (9)$$

The model error of this white-noise approximant, relative to the exact diffusion average, has been estimated separately. It is $O(\kappa)$ relative, plus $O(\Gamma/\Delta\nu)$ from secularisation and the DFS term. This note concerns only the *discretisation* error of the circuit implementing $e^{T\mathcal{L}}$. Following our choice of focus, we treat the white-noise regime in detail and the resolved regime only briefly (section 4.1).

3 Cluster decomposition and the splitting

Partition the spins into geometric clusters $C_1, \dots, C_K$ so that every short distance lies inside a cluster: bonds, geminal pairs and strongly coupled pairs. Examples are CH, CH₂ or CH₃ groups with their carbon, or an aromatic CH pair. Every term is then either intra- or inter-cluster:

$$W_\mu = \underbrace{\sum_c W_\mu^c}_{A_\mu \text{ (all CSA + intra DD)}} + \underbrace{\sum_{c < c'} W_\mu^{cc'}}_{B_\mu \text{ (inter DD)}}, \quad H_{\text{coh}} = \sum_c H_{\text{coh}}^c + \sum_{c < c'} H_{\text{coh}}^{cc'}. \quad (10)$$

The small parameter of the splitting is $\varepsilon_{\text{geo}} = b/a$. Here $a = \max_c \|A_\mu^c\|$ is the intra-cluster scale and $b = \max \|B_\mu^{cc'}\|$ the inter-cluster scale. Since dipolar couplings scale as r^{-3} , a cutoff near $2 \text{ \AA}$ gives $\varepsilon_{\text{geo}} \approx 0.1\text{--}0.3$.

Layers. Build the cluster graph, with an edge c – c' whenever an inter-cluster term exceeds a threshold. Edge-colour it with $\chi \leq \Delta + 1$ colours (Vizing), where Δ is the maximum number of neighbouring clusters. Then:

- $A = \bigoplus_c A^c$ is one layer of commuting blocks. Each block is exponentiated exactly on 2^{n_c} dimensions (16 for $^{13}\text{CH}_3$).
- $B = B^{(1)} + \dots + B^{(\chi)}$, where each colour class is a matching of cluster pairs, hence one layer of commuting blocks.
- Terms far below threshold may be merged into the nearest layer or dropped, with a bias bounded by their norm.

One step (white-noise regime), palindromic at every level:

1. Apply $\mathcal{U}_{\text{coh}}(\delta t/2)$, Strang-split over the same cluster layers (section 6).
2. Apply the kick layer: a palindromic product over channels μ of single-channel kicks, each with an independent Gaussian Ξ_μ .
3. Implement each single-channel kick as a Strang product over cluster layers, driven by one *common* Ξ_μ :

$$K_\mu(X) = e^{-iXA_\mu/2} e^{-iXB_\mu^{(1)}/2} \dots e^{-iXB_\mu^{(\chi)}/2} \dots e^{-iXB_\mu^{(1)}/2} e^{-iXA_\mu/2}. \quad (11)$$

4. Apply $\mathcal{U}_{\text{coh}}(\delta t/2)$.

Two rules are forced by the physics.

- **Common noise across cluster layers.** Independent X for A_μ and B_μ keeps each interaction's auto-relaxation but deletes every cross-correlation between them. This includes DD–CSA cross terms on the same spin and all inter-cluster cross-correlations.
- **Palindromic order inside a kick.** Lie order $e^{-iXA}e^{-iXB}$ generates a spurious Hamiltonian $(\gamma_\mu/2)(-i)[A_\mu, B_\mu]$ at first order in δt . The resulting $O(\Gamma \varepsilon_{\text{geo}} T)$ bias does not vanish as $\delta t \rightarrow 0$.

By locality, $[A^c, B^{c'c''}] \neq 0$ only if $c \in \{c', c''\}$. Every error term below is therefore a sum of contributions supported on one cluster pair and its neighbours.

4 Noise-averaged Strang error

Averaged step. Kicks are independent across steps and channels. The averaged step is therefore a product of averaged single-channel maps, each a mixture of unitaries and hence trace-norm contractive. Since $e^{\delta t \mathcal{L}}$ is also contractive, telescoping gives

$$\|\bar{\rho}_{\text{circ}}(T) - e^{T\mathcal{L}}\rho_0\|_1 \leq \frac{T}{\delta t} \|\bar{\mathcal{S}}(\delta t) - e^{\delta t \mathcal{L}}\|_{1 \rightarrow 1}. \quad (12)$$

Three splittings contribute to the local error.

(i) Coherent $\leftrightarrow$ kick layer. This is a Strang split of $\mathcal{L}_{\text{coh}} = -i \text{ad}_{H_{\text{coh}}}$ against $\mathcal{D} = \sum_\mu \mathcal{D}_\mu$. The local error generator is

$$\delta t^3 \left(-\frac{1}{24} [\mathcal{L}_{\text{coh}}, [\mathcal{L}_{\text{coh}}, \mathcal{D}]] + \frac{1}{12} [\mathcal{D}, [\mathcal{D}, \mathcal{L}_{\text{coh}}]] \right), \quad (13)$$

which is weak order 2, of size $T\delta t^2(J^2\Gamma + J\Gamma^2)$. Offsets drop out because the secular $\mathcal{D}_\mu$ is $U(1)$ -covariant: $[\text{ad}_{H_{\text{off}}}, \mathcal{D}_\mu] = 0$.

(ii) **Between channels.** Each single-channel kick averages exactly to $e^{\delta t \mathcal{D}_\mu}$ by eq. (8). The palindromic product over μ is therefore a Strang split of $\mathcal{D} = \sum_\mu \mathcal{D}_\mu$: weak order 2, of size $T\delta t^2 \Gamma^3$.

(iii) **Inside a kick, across clusters.** For the common scalar X , $K_\mu(X)$ is a Strang formula in the “time” X . The symmetric BCH formula gives

$$K_\mu(X) = \exp\left[-i(XW_\mu - X^3E_\mu) + O(X^5)\right], \quad E_\mu = \frac{1}{12}[B_\mu, [B_\mu, A_\mu]] - \frac{1}{24}[A_\mu, [A_\mu, B_\mu]]. \quad (14)$$

Proposition 1 (Weak order 1 of the intra-kick cluster split). *Let $X \sim \mathcal{N}(0, \gamma_\mu \delta t)$. Then*

$$\mathbb{E}_X[K_\mu \rho K_\mu^\dagger] = e^{\delta t \mathcal{D}_\mu} \rho + \frac{3}{2}(\gamma_\mu \delta t)^2 \{\text{ad}_{W_\mu}, \text{ad}_{E_\mu}\} \rho + O(\delta t^3). \quad (15)$$

Consequently the averaged circuit follows the shifted generator

$$\mathcal{L}_{\text{eff}} = \mathcal{L} + \delta t \mathcal{G} + O(\delta t^2), \quad \mathcal{G} = \frac{3}{2} \sum_\mu \gamma_\mu^2 \{\text{ad}_{W_\mu}, \text{ad}_{E_\mu}\}, \quad \|\Delta \bar{\rho}(T)\|_1 \leq T \delta t \|\mathcal{G}\|. \quad (16)$$

Proof. Write the channel as $e^{X\mathcal{A} + X^3\mathcal{C} + O(X^5)}$, with $\mathcal{A} = -i \text{ad}_{W_\mu}$ and $\mathcal{C} = i \text{ad}_{E_\mu}$. Odd moments of X vanish. At order X^4 , the terms beyond the exact Gaussian average $e^{\gamma \delta t \mathcal{A}^2/2}$ are $\frac{1}{2}X^4(\mathcal{A}\mathcal{C} + \mathcal{C}\mathcal{A})$. Using $\mathbb{E}[X^4] = 3(\gamma_\mu \delta t)^2$ and $\mathcal{A}\mathcal{C} = \text{ad}_{W_\mu} \text{ad}_{E_\mu}$ gives eq. (15). Summing the $T/\delta t$ local errors gives the generator shift. $\square$

The correction $\mathcal{G}$ is a correlated-dissipator term between W_μ and its own splitting error. Its size is $\|\mathcal{G}\| \lesssim 12 \sum_\mu \gamma_\mu^2 \|W_\mu\| \|E_\mu\| \sim \Gamma^2 \varepsilon_{\text{geo}}$. It is linear in the inter-cluster coupling b , because $\|[A, [A, B]]\| \sim a^2 b$ dominates. Table 2 compares implementation options; section 5 gives the minimal order-2 kick.

Kick implementation	Local averaged error	Global error	Layers per kick
Lie, $e^{-iXA}e^{-iXB}$	$(\gamma \delta t/2)[A, B]$	$O(T\Gamma \varepsilon_{\text{geo}})$, no $\delta t \rightarrow 0$ limit	2
Strang in X , eq. (11)	$\frac{3}{2}(\gamma \delta t)^2 \{\text{ad}_W, \text{ad}_E\}$	$T\delta t \ \mathcal{G}\ $, weak order 1	$2\chi + 1$
Triple jump in X (section 5)	$O((\gamma \delta t)^3)$	$O(T\delta t^2 \Gamma^3 \varepsilon_{\text{geo}})$, weak order 2	$6\chi + 1$
Suzuki S_4 in X (5 stages)	$O((\gamma \delta t)^3)$	weak order 2	$10\chi + 1$
Strang + Richardson	0 at $O(\delta t)$	weak order 2	$\sim 1.5 \times$ gates, $\times 5$ variance

Table 2: Options for the cluster split inside a single-channel kick.

4.1 Resolved regime

Per trajectory, Strang over the same cluster layers with R at step midpoints has local error $\delta t^3 E(R_k)$. Its fluctuating part averages out over τ_c . What accumulates is the Haar mean $\langle E(R) \rangle$, which is nonzero through the third moment of $D^{(2)}$. This gives $T\delta t^2 \|\langle E \rangle\| \sim T\delta t^2 h^3 \varepsilon_{\text{geo}}$, about 10^{-7} at $\delta t = \tau_c/10$.

In this regime the orientation discretisation dominates instead. With exact increments the relative rate error is $\delta t^2/(12\tau_c^2)$; with naive Gaussian increments it is $O(\delta t)$. The Zeeman term should be handled in the interaction picture. Since H_Z is a sum of single-spin terms, this preserves cluster locality. Putting H_Z in a Trotter layer instead costs $T\delta t^2 \omega_0^2 b \sim O(10)$ at $\delta t = \tau_c/10$.

5 Minimal order-2 kick: triple jump in X

The simplest kick with weak order 2 is a three-stage triple jump in the noise variable. It uses fixed weights and the same X in every sub-kick:

$$\begin{aligned} K_\mu^{(2)}(X) &= K_\mu(w_1 X) K_\mu(w_0 X) K_\mu(w_1 X), \\ w_1 &= \frac{1}{2 - 2^{1/3}} \approx 1.3512, \quad w_0 = 1 - 2w_1 \approx -1.7024. \end{aligned} \tag{17}$$

$X \sim \mathcal{N}(0, \gamma_\mu \delta t)$ is drawn as before, and each $K_\mu(\cdot)$ is the Strang-in- X product eq. (11). The global error falls from $T\delta t \|\mathcal{G}\|$ to $O(T\delta t^2)$.

Proposition 2 (Order condition). *Consider a palindromic composition of Strang kicks with sub-steps $c_i(X)$, $\sum_i c_i = X$. Its exponent is $-i(XW_\mu - c_3(X)E_\mu) + O(X^5)$, with $c_3 = \sum_i c_i^3$; even-order BCH terms cancel by symmetry. Then*

$$\mathbb{E}[K\rho K^\dagger] - e^{\delta t \mathcal{D}_\mu} \rho = \frac{1}{2} \mathbb{E}[X c_3(X)] \{\text{ad}_{W_\mu}, \text{ad}_{E_\mu}\} \rho + O(\delta t^3). \tag{18}$$

Hence weak order 2 holds if and only if $\mathbb{E}[X c_3(X)] = 0$. The cubic coefficient must be orthogonal to X ; it need not vanish.

The plain kick has $c_3 = X^3$, so $\mathbb{E}[X^4] = 3(\gamma\delta t)^2$, recovering proposition 1.

Nothing cheaper works.

- *Changing the law of X .* The exact part needs $\mathbb{E}[X^2] = \gamma\delta t$ and $\mathbb{E}[X^4] = 3(\gamma\delta t)^2$ to reproduce $e^{\delta t \mathcal{D}_\mu}$ at $O(\delta t^2)$. This fixes $\mathbb{E}[X \cdot X^3] = 3(\gamma\delta t)^2 \neq 0$.
- *Two sub-kicks $c_1 + c_2 = X$,* even with extra independent randomness. Then $c_1^3 + c_2^3 = X(X^2/4 + 3\eta^2)$ with $\eta = (c_1 - c_2)/2$. This always has the sign of X , so $\mathbb{E}[X c_3] > 0$.
- *Random ordering* (ABA vs BAB). A mixture cancels $[B, [B, A]]$ only at weight 1/3 and $[A, [A, B]]$ only at weight 2/3, so no single mixture removes both.

Three symmetric stages with one backward stage are therefore the minimum. With deterministic weights $c_i = w_i X$, the condition $\mathbb{E}[X c_3] = 0$ reduces to $w_1(1 - w_1)^2 = 1/6$. This is exactly Yoshida's triple-jump condition, so c_3 vanishes identically.

Noise-adapted alternative. One can instead keep $c_3 \neq 0$ and make it the Hermite polynomial $c_3 = X^3 - 3\gamma\delta t X$, which is orthogonal to X . Use sub-steps $(p, X - 2p, p)$, with $p(X)$ the smallest root, of the same sign as X , of

$$p(X - p)^2 = \frac{1}{2} \gamma_\mu \delta t X. \tag{19}$$

This is equivalent to kicking with $W_\mu + 3\gamma_\mu \delta t E_\mu$, i.e. pre-compensating the error. The sub-steps stay close to $(0, X, 0)$, but a cubic must be solved for each sample, and the test system shows no accuracy gain over eq. (17).

Cost. After merging adjacent A half-layers, a kick with χ inter-cluster colours takes $6\chi + 1$ exponential layers. This compares with $2\chi + 1$ for Strang in X and $10\chi + 1$ for the five-stage Suzuki S_4 . Sub-kick angles grow to $1.7|X|||W|| \sim 1.7\sqrt{\Gamma\delta t} \ll 1$.

6 Commutator bounds per term class

Cluster locality makes every nested commutator a sum over cluster pairs. Per channel μ , write $a_c = \|A_\mu^c\|$, $b_{cc'} = \|B_\mu^{cc'}\|$, and $\beta_c = \sum_{c'} b_{cc'}$ for the total inter-cluster coupling of cluster c . Then

$$\|[A_\mu, [A_\mu, B_\mu]]\| \leq 4 \sum_{(c,c')} (a_c + a_{c'})^2 b_{cc'}, \quad \|[B_\mu, [B_\mu, A_\mu]]\| \leq 8\bar{\beta} \sum_c a_c \beta_c, \quad \bar{\beta} = \max_c \beta_c, \quad (20)$$

$$\|E_\mu\| \leq \frac{1}{6} \sum_{(c,c')} (a_c + a_{c'})^2 b_{cc'} + \frac{2}{3}\bar{\beta} \sum_c a_c \beta_c. \quad (21)$$

In $\{\text{ad}_W, \text{ad}_E\}$ only the part of W_μ overlapping the support of E_μ contributes, so $\|W_\mu\|$ may be replaced by its local norm $\approx 2a$. When $b \ll a$ the first term dominates, and each cluster pair contributes $\approx 16\gamma^2 a^3 b$ to $\|\mathcal{G}\|$. Hence

$$\|\mathcal{G}\| \leq 16 \sum_{(c,c')} \Gamma_c^2 \varepsilon_{cc'}, \quad \Gamma_c = \gamma a_c^2, \quad \varepsilon_{cc'} = b_{cc'}/a_c. \quad (22)$$

Term pair	Commutes?	Enters	Scale
CSA _i –CSA _j ; CSA _i –DD _{jk} , $i \notin \{j, k\}$	yes	—	0
CSA _i –inter-cluster DD _{ij}	no	$[A, [A, B]]$ in E_μ	$\delta_i^2 b_{ij}$
Intra DD–inter DD sharing a spin	no	$[A, [A, B]]$ in E_μ	$a_{\text{DD}}^2 b$
Inter DD of different colours sharing a spin	no	$[B, [B, A]]$, inter-colour	ab^2
Inter DD of the same colour	yes (disjoint)	—	0
Weakly coupled (secular) J , off-sets	yes, mutually	only in (i) vs $\mathcal{D}$	$J^2\Gamma + J\Gamma^2$
Strong inter-cluster J (flip-flop)	no, with offsets	coherent Strang	$\Delta_{ij}^2 J_{ij}/24$ per bond
H_{DFS} pieces	overlapping supports	coherent layers	$\Gamma\omega\tau_c \times J$

Table 3: Which term pairs fail to commute, and where they enter the error.

These bounds have three consequences for choosing clusters.

- **Put strongly coupled pairs inside a cluster.** The coherent part is then one exactly solvable diagonal layer (offsets and secular J) plus intra-cluster blocks. Its Trotter error vanishes except for the type-(i) term.
- **Put every spin’s CSA in A .** It is single-spin, so it never costs a layer. It enters only through $[\text{CSA}_i, \text{DD}_{ij}]$ on inter-cluster bonds.
- **Minimise $\sum_{(c,c')} a_c^2 b_{cc'}$, not the number of clusters.** Since $b \sim r^{-3}$, the error is dominated by the closest inter-cluster contacts: vicinal ^1H – ^1H at 2.3–2.5 Å, and ^{13}C – ^{13}C and ^{13}C – ^1H across a bond. Merging two clusters removes $b_{cc'}$ at the cost of a $2^{n_c+n_{c'}}$ -dimensional block.

7 Error budget and numerical check

The total discretisation error of the averaged circuit over time T is

$$\varepsilon(T) \approx \underbrace{T\delta t \|\mathcal{G}\|}_{\text{kick cluster split}} + \underbrace{c_1 T\delta t^2 (J^2\Gamma + J\Gamma^2)}_{\text{coherent} \leftrightarrow \text{kicks}} + \underbrace{c_2 T\delta t^2 \Gamma^3}_{\text{channel ordering}} + \underbrace{T\delta t^2 \sum \Delta_{ij}^2 J_{ij}/24}_{\text{strong inter-cluster } J} + \frac{\|O\|}{\sqrt{M}}. \quad (23)$$

The first term is extensive in the number of cluster contacts for the full state. For a local observable, only contacts inside its light cone count. With the triple jump (or Richardson extrapolation), the first term becomes $O(T\delta t^2\Gamma^3\varepsilon_{\text{geo}})$ and the whole budget is weak order 2.

Test system. A $^{13}\text{C}_2$ -labelled trans HC=CH fragment, with two clusters $\{\text{C}_1, \text{H}_1\}$ and $\{\text{C}_2, \text{H}_2\}$. Bond lengths are C=C 1.34 Å and C–H 1.08 Å. The ^{13}C CSA is 150 ppm and the ^1H CSA 5 ppm, with random principal axes, at 14.1 T with $\tau_c = 100$ ps. All 25 zero-frequency channels (a, b) are kept. Exact averaged channels were computed by 40-point Gauss–Hermite quadrature on the 256-dimensional Liouville space.

In this system $a \approx 2.4 \times 10^5 \text{ s}^{-1}$, $b \approx 3.5 \times 10^4 \text{ s}^{-1}$, $\varepsilon_{\text{geo}} \approx 0.14$, and $\Gamma \approx 0.66 \text{ s}^{-1}$ per channel. Exactly, $\sum_{\mu} \gamma_{\mu}^2 \|\frac{3}{2}\{\text{ad}_W, \text{ad}_E\}\| = 0.84 \text{ s}^{-2}$; the crude norm bound gives 1.75 s^{-2} .

Check	$\delta t = 1 \text{ ms}$	0.3 ms	0.1 ms	Order
Relative residual of eq. (15) (one channel)	2.1×10^{-3}	6×10^{-4}	2×10^{-4}	residual $\propto \delta t$
Strang-in- X kicks, max obs. error, $T = 1 \text{ s}$	3.7×10^{-7}	1.1×10^{-7}	3.7×10^{-8}	weak 1
Triple-jump kicks, same	4.3×10^{-10}	—	5.6×10^{-12}	weak 2
Hermite triple jump, same	2.1×10^{-10}	—	6.0×10^{-12}	weak 2
Suzuki S_4 kicks, same	7×10^{-12}	6×10^{-13}	round-off	weak 2
Lie kicks, local error (one channel)	2.4×10^{-4}	7.3×10^{-5}	2.4×10^{-5}	global $O(1)$
Coherent $\leftrightarrow$ kick, secular $J \leq 155 \text{ Hz}$, I_z , $T = 1 \text{ s}$	3.4×10^{-9}	—	3.3×10^{-11}	weak 2
Same, transverse I_x , $T = 50 \text{ ms}$	1.9×10^{-5}	1.7×10^{-6}	1.8×10^{-7}	weak 2

Table 4: Numerical checks on the HC=CH fragment. Observables are $\langle I_z \rangle$, $\langle 2I_z S_z \rangle$, $\langle I_x \rangle$ and $\langle 2I_x S_z \rangle$ on the C and H spins, normalised by $\text{tr}(O^2)$.

Kick	Local, $\delta t = 1 \text{ ms}$	Local, 0.3 ms	Global, 1 ms	Global, 0.1 ms
Strang in X	1.5×10^{-7}	1.3×10^{-8}	3.7×10^{-7}	3.7×10^{-8}
Triple jump, fixed w	1.2×10^{-10}	3.1×10^{-12}	4.3×10^{-10}	5.6×10^{-12}
Triple jump, Hermite $p(X)$	1.4×10^{-10}	3.9×10^{-12}	2.1×10^{-10}	6.0×10^{-12}
Suzuki S_4 , 5 stages	2.2×10^{-11}	5.8×10^{-13}	—	—

Table 5: Order-2 kicks: local error in the worst channel, and global error at $T = 1 \text{ s}$ with all 25 channels. The local error scales as δt^3 for all three order-2 kicks; the quadrature confirms $\mathbb{E}[Xc_3] = 0$ for the Hermite version to 10^{-15} .

The norm bound $T\delta t\|\mathcal{G}\|$ predicts 8.4×10^{-4} at $\delta t = 1 \text{ ms}$, while the measured magnetisation error is 3.7×10^{-7} . The bound is therefore about $2000\times$ loose in this case. It is dominated by high-order coherences that low-rank observables barely feel, and it ignores dissipative contraction.

Remark 1. The check uses only zero-frequency channels (a $J(0)$ white-noise model). Frequency-resolved channels change γ_{μ} and split W_{μ} into Bohr components. They do not change the cluster structure of E_{μ} , so the orders and the $\gamma^2 a^3 b$ scaling carry over.

Step size. Take $\varepsilon = 10^{-3}$, $T = 1 \text{ s}$, about 10 cluster contacts, $\Gamma \sim 1 \text{ s}^{-1}$ and $\varepsilon_{\text{geo}} \sim 0.2$. Then:

In practice δt is set by resolving the J evolution ($J\delta t \lesssim 0.1\text{--}1$). This gives $10^3\text{--}10^4$ steps per second, against $\sim 10^{11}$ in the resolved regime.

Constraint	Rigorous (norm bound)	Realistic (test system)
Kick cluster split, Strang in X	$\delta t \leq 3 \times 10^{-5} \text{ s}$	$\delta t \lesssim 10^{-1} \text{ s}$ (not binding)
Coherent $\leftrightarrow$ kick, $^1J_{\text{CH}} = 140 \text{ Hz}$	$\delta t \leq 10^{-4} \text{ s}$	$\delta t \lesssim 10^{-3} \text{ s}$
Kick-formula validity	$\Gamma \delta t \ll 1, \tau_c \ll \delta t$	—

Table 6: Step-size constraints for $\varepsilon = 10^{-3}$, $T = 1 \text{ s}$.

8 Summary and recommended splitting

With a cluster-aware splitting and the Strang-in- X kick, the second-order circuit has global error $T\delta t\|\mathcal{G}\| + O(T\delta t^2)$. Here $\|\mathcal{G}\| \sim \Gamma^2\varepsilon_{\text{geo}}$ per cluster contact, and this term is the only obstruction to full weak order 2. The recommended construction is:

1. Choose clusters that contain every bond, geminal pair and strongly coupled pair, and minimise $\sum a_c^2 b_{cc'}$ over the closest inter-cluster contacts.
2. Coherent part: one exact diagonal layer (offsets and secular J), then intra-cluster blocks, then edge-coloured inter-cluster blocks, all palindromic.
3. Kicks: one independent Gaussian Ξ_μ per channel (a, b, ν) with $\tilde{J}(\nu)$ -weighted variance, in palindromic channel order.
4. Inside each kick: the same Ξ_μ on all cluster layers, in Strang order $A/2 \cdot B \cdot A/2$. Never use independent noise per cluster, which deletes cross-correlations, nor Lie order, which leaves an $O(\Gamma\varepsilon_{\text{geo}}T)$ bias.
5. For weak order 2, use the fixed-weight triple jump in X , eq. (17), with $6\chi + 1$ layers per kick. In the test system Strang in X is already at 10^{-7} , so this is needed only for high-precision targets or many contacts.
6. Use the resolved regime only to validate the white-noise model, with H_Z in the interaction picture and exact $SO(3)$ increments; the cluster-split error is negligible there.

Open items.

- Repeat the check with frequency-resolved channels on a real molecule, e.g. a ^{13}C -labelled amino acid with a CH_3 cluster.
- Derive the X^6 coefficient of the order-2 kicks explicitly.
- Find an observable-weighted bound that tracks the measured error, which is about $2000\times$ below the norm bound.